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10 CFR 50.73

July 17, 2023
Serial: RA-23-0138

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit No. 2
Renewed Facility Operating License No. DPR-62
Docket No. 50-324
Licensee Event Report 2-2023-001

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, is submitting the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Mr. Mark DeWire, Manager – Nuclear Support Services, at (910) 832-6641.

Sincerely,

A handwritten signature in black ink, appearing to read "John A. Krakuszeski", written in a cursive style.

John A. Krakuszeski

SBY/sby

Enclosure: Licensee Event Report

cc (with enclosure):

Ms. Laura Dudes, NRC Regional Administrator, Region II
Mr. Luke Haeg, NRC Project Manager
Mr. Gale Smith, NRC Senior Resident Inspector



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Brunswick Steam Electric Plant (BSEP), Unit 2

☒ 050
☐ 052

2. Docket Number

00324

3. Page

1 OF 4

4. Title

Two Main Steam Line Safety Relief Valves Inoperable

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
05	23	2023	2023	- 001 -	00	07	17	2023	Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Mark DeWire, Manager – Nuclear Support Services

Phone Number (Include area code)

(910) 832-6641

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SB	RV	T020	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On May 23, 2023, BSEP received test results for the eleven main steam line safety relief valves (SRVs) removed from Unit 2 during the spring 2023 refuel outage (RFO). Two of the eleven valves were found to have as found pilot valve lift setpoints outside the +/-3 percent tolerance required by Technical Specification (TS) 3.4.3, which requires ten of the eleven installed SRVs to be operable. All eleven of the SRV pilot valves were replaced with certified spares before Unit 2 startup from the spring 2023 RFO. At the time the test results were received, Unit 2 was operating in Mode 1 at approximately 100% power.

Corrosion bonding of the SRV pilot disc to the pilot seat likely caused one of the out of tolerance as-found lift setpoints. The other out of tolerance lift setpoint was likely a result of a combination of degraded pilot valve assembly sub-components.

Since less than ten SRVs were operable and the failure modes likely arose over a period of time during Unit 2 operation, this event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's TS.

There was no impact on the health and safety of the public or plant personnel.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Brunswick Steam Electric Plant (BSEP), Unit 2	☒ 050	2. DOCKET NUMBER 00324	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

Background*Initial Conditions*

At the time the test results were received, Unit 2 was in Mode 1 (i.e., Power Operation), at approximately 100 percent rated thermal power.

Reportability Criteria

Unit 2 Technical Specification (TS) 3.4.3 requires at least ten of eleven main steam [SB] line safety relief valves (SRVs) to be operable. Per Surveillance Requirement (SR) 3.4.3.1, each valve is required to open within +/- 3 percent of its opening setpoint. As-found testing of the valves indicated that two of the valves had lift setpoints outside this tolerance. Since the failure modes likely arose over a period of time during Unit 2 operation, the plant likely operated in a condition which is prohibited by the TS, that is, with fewer than the required number of SRVs having lift setpoints within the +/-3 percent tolerance. Therefore, the condition is being reported per 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's TS.

Event Description

During the spring 2023 Unit 2 refueling outage, all eleven pilot valve assemblies in the SRVs were replaced with certified spares. The removed SRV pilot valves were sent to NTS Labs, LLC, to determine the as-found set pressure. On May 23, 2023, test results were reported to BSEP. The test results showed that two of the eleven valves actuated at pressures outside of the +/-3 percent tolerance allowed by TS SR 3.4.3.1. Specifically, pilot valve with Serial Number (SN) 1278 lifted 3.8 percent above the nameplate set pressure, and pilot valve with SN 1223 lifted 5 percent below the nameplate set pressure.

Event Cause

For the SN 1278 pilot valve, corrosion bonding of the SRV pilot disc to the pilot seat likely caused the +3.8 percent out of tolerance as-found lift setpoint. Corrosion bonding between the pilot disc and seat is an inherent problem with the Target Rock two-stage SRV design used at BSEP that has been effectively mitigated by coating each pilot disc surface with Platinum. However, due to the complexities associated with the Platinum coating process and the harsh environment in which the SRVs operate, corrosion bonding may occur periodically.

For the SN 1223 pilot valve, the -5 percent out of tolerance lift setpoint was likely a result of a combination of degraded pilot valve assembly sub-components which were noted during disassembly inspections. Minor steam cutting was noted on the pilot disc which likely resulted from slight seat leakage. Minor scoring was noted between the guide assembly and the seal bellows rod assembly indicating side-loading of the spring. In addition, the set pressure spring was not a matched set with the spring follower which likely resulted in the spring repositioning after being lifted and contributed to the side-loading of the spring. Minor steam cutting and minor scoring are considered normal wear. It should be noted that the mismatched set pressure spring assembly produced satisfactory results during the

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NARRATIVE

certification process and no leakage was observed during said certification. Therefore, it is unlikely that any one of these conditions alone would result in the out of tolerance lift setpoint; however, the combined effect of the degraded components likely resulted in the -5 percent out of tolerance lift setpoint.

Safety Assessment

The purpose of the SRVs is to provide overpressure protection for the reactor coolant system. The as-found condition of the Unit 2 SRVs was compared to cases analyzed in the Brunswick Unit 2 Cycle 25 Reload Safety Analysis Report (RSAR). Each as-found SRV was bounded by the opening pressures assumed in the analysis; therefore, the overall RSAR analysis bounds the overall as-found condition of the SRVs.

The results from the RSAR analysis demonstrate that the pressure limits are not exceeded for the overpressurization events. Therefore, it is concluded that the Unit 2 Cycle 25 SRVs always remained capable of performing their safety function of preventing overpressurization of the reactor coolant system.

Based on the foregoing analysis, it is concluded that this event had no adverse impact on nuclear safety.

Corrective Actions

All eleven of the SRV pilot valves were replaced with certified spares before Unit 2 startup from the spring 2023 refueling outage. Also, it was confirmed that all other valves removed from Unit 2 during the spring 2023 refueling outage have matching spring sets and rebuild paperwork for all valves in operation in Unit 1 and Unit 2 document matching spring sets.

In addition to the aforementioned completed corrective actions, the SRV pilot valve assemblies with lift setpoints outside +/- 3 percent will be rebuilt and certified for use. The rebuild process restores the pilot disc surface finish to a satisfactory condition and recoats the pilot disc with Platinum to prevent corrosion bonding. The rebuild process will also ensure that degraded pilot valve assembly sub-components are restored to acceptable conditions, and the set pressure spring, retainer, and follower are a matched set. The rebuild campaign is scheduled to complete prior to the end of 2023.

Any changes to corrective actions or completion schedules will be made in accordance with the site's corrective action program.

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NARRATIVE**Previous Similar Events**

A review of LERs for the past three years identified the following previous similar event associated with SRVs with as-found lift setpoints outside of TS allowable limits.

- LER 1-2020-004, dated September 29, 2020, reported Unit 1 operation prohibited by TSs due to three of the eleven valves found with lift setpoints of their pilot valves outside the +/-3 percent tolerance required by TS SR 3.4.3.1.

The direct cause of this event was corrosion bonding. The root cause was attributed to the procedures in place at the time these valves were refurbished lacking sufficient detail to ensure consistent surface preparation and proper quality checks of surface condition prior to Platinum coating. The procedures were revised and utilized during the rebuild process for the subject pilot valves.

Commitments

No regulatory commitments are contained in this report.